

CUSTOMER FEATURE GUIDE

GuestKit

Offline VM intelligence and migration assurance.

Inspect, score, and repair QCOW2, VMDK, and RAW disk images without ever powering the VM on.

by ZyvorAI Labs

What is GuestKit?

GuestKit is a pure-Rust control plane that reads a virtual machine's disk image offline, builds a normalized evidence snapshot, and answers the two questions that decide every migration: will it boot, and what must change before cutover. It ships as a scriptable CLI (guestkit), a carbon-themed TUI (guestctl), Python bindings, and a self-hosted web platform with KubeVirt integration - all sharing one engine, with no libguestfs appliance to install.

70+

CLI subcommands

6

disk formats read

0

libguestfs appliances needed

8

migration targets scored

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Offline Disk Inspection

Read the full guest OS from a cold disk image - no boot, no agent, no appliance.



One-command inspect

guestkit inspect surfaces OS, distro, version, hostname, architecture, and init system from any supported disk image in a single pass.

Know exactly what a VM is before you touch it.



Six disk formats, auto-detected

Reads QCOW2, VMDK, VHD, VHDX, VDI, and RAW/IMG images, choosing loop devices or qemu-nbd automatically.

Point it at whatever your hypervisor exported - it just opens.



Guest OS detection

Fingerprints Linux distributions (Fedora, Ubuntu, Debian, RHEL, CentOS, SUSE and more) plus Windows from on-disk signals.

Accurate identity without a running kernel.



Deep system enumeration

Extracts packages, kernels, users, SSH config, services, timers, network, DNS, LVM, fstab, runtimes, containers, certificates, and cloud-init state.

A complete inventory of the machine from bytes on disk.



Windows signal parsing

Reads SAM/SECURITY registry hives to detect BitLocker, domain join, RDP, and driver gaps for Windows guests.

See the Windows-specific blockers Linux tools miss.



Pure-Rust engine, no libguestfs

Partition tables, filesystem signatures, and evidence schema are parsed in Rust; only host NBD/loop is used for mount.

No guestfish appliance, no fragile daemon - fewer moving parts.

Inspection is always read-only: your source image is never modified.

Migration Assurance

Score boot readiness and generate hypervisor-aware fix plans before you cut over.

Target	Boot analysis	Migration rules applied
kvm / proxmox / qemu	Proxmox/KVM	VirtIO, virtio-scsi/net, VMware Tools to qemu-ga
aws / azure / gcp / cloud	Cloud	cloud-init datasource, BYOL licensing
hyperv	Hyper-V	Hyper-V-specific boot checks



Boot assurance score

guestkit doctor predicts first-boot success on a target hypervisor with a 0-100 score, ranked blockers, and warnings.

Answer 'will it boot?' before the weekend, not during it.



Root-cause --explain

An inference engine traces each blocker back through a causal chain so you see why a VM would fail to boot.

Fix the cause, not the symptom.



Migration score + checklist

guestkit migrate-plan applies target-specific rules for VirtIO drivers, cloud-init, VMware Tools removal, BitLocker, and SELinux relabel across eight targets.

A tailored cutover checklist per destination platform.



CI boot gate

--fail-below sets an exit-code threshold so pipelines block any image that scores under your bar, JSON still emitted.

Golden images that regress never reach production.



Policy-as-code

guestkit policy check evaluates an expression DSL over evidence fields (e.g. bootability.score >= 80) or built-in CIS benchmarks.

Codify sign-off criteria your whole team can trust.



Forensic diff

guestkit forensic-diff compares two snapshots for config drift, suspicious persistence, and ransomware indicators.

Prove what changed between golden and drifted.

Fix Plans & Repair

Turn findings into reviewable, reversible, executable remediation - not blind edits.



Reviewable fix plans

Findings become a structured plan of operations (file edits, package installs, service ops, SELinux, registry edits) that you preview before anything runs.

[See every change before it happens.](#)



Transactional boot repair

guestkit repair --fix boot converts doctor blockers into a plan, applies it with backups, then re-scores to show the delta.

[Fix boot blockers offline and prove the score improved.](#)



Export to bash & Ansible

Plans export as executable shell scripts, Ansible playbooks, JSON, or YAML for change control and runbooks.

[Hand ops a runbook your CAB can approve.](#)



Backup & rollback

guestkit plan apply creates timestamped backups; plan rollback restores prior state, with dependency ordering and dry-run.

[Every change has an undo button.](#)



Automated hardening

guestkit harden generates security-profile fixes for SSH, firewall, SELinux/AppArmor, and account posture.

[Ship hardened images without hand-editing configs.](#)



Offline agent injection

repair --inject-agent writes a guest agent binary into the disk during migration prep, no boot required.

[The VM comes up already instrumented.](#)

Security teams generate plans; ops teams apply them - full separation of duties, in version control.

Security & Threat Analysis

Audit posture, hunt for compromise, and prove compliance - all from the offline disk.



Security profile audit

guestkit inspect --profile security scores SSH exposure, UID-0 users, firewall, SELinux/AppArmor, and kernel into a risk level.

[A ranked risk verdict per VM in seconds.](#)



Secret & credential scan

guestkit secrets sweeps the disk for exposed credentials and keys.

[Catch leaked secrets before an image ships.](#)



Malware & rootkit detection

guestkit malware scans for rootkits and known-bad artifacts offline, where in-guest malware can't hide from the scanner.

[Inspect a suspect image without executing it.](#)



CVE & patch analysis

guestkit cve maps installed packages to known vulnerabilities and missing security patches.

[See the VM's exposure without a live agent.](#)



Compliance checking

guestkit compliance and audit evaluate images against security standards with detailed reporting.

[Turn every VM into an audit artifact.](#)



Threat hunting & IOC

guestkit threat-intel, hunt, and anomaly correlate indicators, detect anomalies, and surface suspicious persistence offline.

[Forensic triage on a dead disk, safely.](#)



Forensic timeline & reconstruction

guestkit timeline and reconstruct build an incident timeline from multiple on-disk sources and visualize the attack path.

[Rebuild what happened without booting the evidence.](#)

Inventory & Reporting

Produce SBOMs, license reports, and shareable documents from any image.



SBOM generation

guestkit sbom emits a software bill of materials in SPDX or CycloneDX from the guest package set.

[Supply-chain inventory for every VM you run.](#)



License compliance

guestkit licenses inventories package licenses across the disk for compliance review.

[Know your license exposure before an audit asks.](#)



Self-contained HTML reports

--export html builds an interactive, collapsible, print-friendly report with all CSS and JS embedded.

[Email a single file to any stakeholder.](#)



Git-friendly Markdown

--export markdown produces version-controllable inventory documents for VM-configuration history.

[Track infrastructure drift in your docs repo.](#)



Machine-readable output

Most commands accept -o json or -o yaml for automation, monitoring, and jq/yq pipelines.

[Wire GuestKit straight into your tooling.](#)



Fleet posture analysis

guestkit fleet analyze scans a directory of images, clusters identical OS fingerprints, and flags snowflakes and low-score blockers.

[See fleet-wide drift at a glance.](#)

--output (json/yaml/text) and --export (html/markdown) are mutually exclusive - run twice for both. PDF is produced via external tools (wkhtmltopdf, headless browser).

Planning & Optimization

Reverse-engineer infrastructure-as-code, model cloud cost, and map dependencies from a disk.



Infrastructure-as-code blueprints

guestkit blueprint generates Terraform, Ansible, Kubernetes, or Docker Compose definitions from what it finds on the image.

[Recreate a legacy VM as code you can redeploy.](#)



Cloud cost analysis

guestkit cost profiles the workload and estimates run cost plus savings opportunities across AWS, Azure, and GCP.

[Price the migration before you commit to a cloud.](#)



Dependency graph

guestkit dependencies builds a package dependency graph with conflict, circular-dependency, and impact analysis.

[Understand blast radius before you change anything.](#)



Disk format conversion

guestkit convert transcodes images between the six supported formats using qemu-img.

[Reformat once, migrate anywhere.](#)



Smart recommendations

guestkit recommend and predict surface tuning and remediation guidance grounded in the evidence snapshot.

[Actionable next steps, not just raw data.](#)



Performance profile

--profile performance flags swappiness, I/O scheduler, mount options, and network tuning opportunities.

[Baseline and tune before cutover.](#)

Interactive Workspaces

A carbon TUI, a file explorer, and a shell for hands-on offline investigation.



Carbon-themed TUI

guestctl tui opens a k9s-style dashboard with grouped views, vim keys, a command palette, and glass/transparency themes.

[Explore a VM visually without leaving the terminal.](#)



Assurance view parity

The TUI Assurance tab runs doctor, cycles targets (kvm/proxmox/aws), previews fix plans, and exports YAML - reusing the CLI engine on one mount.

[Full assurance workflow, keyboard-driven.](#)



Interactive file explorer

guestkit explore browses partitions and files in place with view, info, filter, sort, and hidden-file toggles.

[Grep-free spelunking through a cold disk.](#)



Guest shell & REPL

guestkit shell and interactive give ls/cat/grep/find over the mounted image plus a scriptable session.

[Familiar Unix muscle memory on any VM.](#)



Fleet & compare modes

--fleet browses a directory of images with a sidebar; --compare diffs two VMs side by side in the dashboard.

[Spot the odd VM out across a set.](#)



Global search & jump

Cross-view search finds packages, boot blockers, and migration items; a grouped jump menu navigates every view.

[Find any signal without knowing which tab holds it.](#)



AI copilot Q&A

guestkit ai answers natural-language questions grounded in the evidence snapshot, with pluggable LLM backends (OpenAI, Anthropic, xAI, or local Ollama).

[Ask a VM what's wrong and get an evidence-backed answer.](#)

Guest Agent & Live Control

Run inside the guest - or reach it host-mediated - even when there's no guest network.



In-guest agent

guestkit agent runs like qemu-guest-agent over virtio-serial, reusing the same evidence and fix-plan schema as offline mode.

[One model for cold-disk and live guests.](#)



Transport ladder

The Guest Control Fabric auto-selects the best path per VM - virtio-serial, QGA exec, QGA builtin, push cache, offline disk, or console.

[Guest control that never depends on guest networking.](#)



Snapshot quiesce

Freeze and thaw guest filesystems (fsfreeze) for application-consistent snapshots, plus soft reboot and graceful shutdown.

[Clean snapshots without crash-consistency risk.](#)



Live remediation with approval

Restart failed units, collect support bundles, and run fix plans - policy-gated with JIT approval workflows.

[Safe, audited guest actions at fleet scale.](#)



mTLS & signed updates

Agents bootstrap client certs, push heartbeats over mTLS, and self-update from Ed25519-signed, SHA256-verified bundles.

[A hardened, tamper-evident guest agent.](#)



Deep Linux health

Component scores for boot, systemd, network, DNS, storage, and security via systemd D-Bus, journald, /proc, and PSI pressure.

[Root-cause the failed unit from journal correlation.](#)

Deep guest intelligence targets Linux; Windows uses virtio-win and a scheduled-task updater today, with a native agent MSI scaffolded.

KubeVirt & Platform

Boot-inspect stopped VMs in-cluster and drive it all from a self-hosted web console.



Offline boot-inspect for stopped VMs

zyvor-api resolves a stopped VM's root PVC and runs guestkit boot-inspect, returning fstab validity, bootloader, and cloud-init state.

[Assurance for halted VMs without booting them.](#)



Zeus VM Tools

A Kubernetes-native guest agent with cloud-init, QGA, ISO, and airgap install paths plus VMToolsPolicy auto-install/upgrade reconciliation.

[The VMware Tools equivalent for KubeVirt.](#)



Web console

Self-hosted zyvor-ui + zyvor-api + guestkit-worker ship as public GHCR images and a Helm chart, backed by a Redis job queue.

[A team-facing UI over the same engine.](#)



Python bindings

hypersdk-guestkit on PyPI exposes a libguestfs-style Guestfs API (100+ methods) for programmatic inspection.

[Automate disk inspection from Python.](#)



hyper2kvm pipeline

Pairs with hyper2kvm for VMware-to-KVM conversion, sitting in the wider HyperSDK to GuestKit to v9s to PacketWolf flow.

[One assurance gate inside a full migration pipeline.](#)



Pluggable auth

The web stack supports JWT, local login, and OIDC/SAML hooks with JWKS-verified ID tokens.

[Wire the console into your existing identity.](#)

In-cluster boot-inspect needs a privileged pod or node disk access plus get/list RBAC on VMs, VMIs, PVCs, and PVs. A running VM returns VM-spec heuristics - offline disk access is for stopped VMs.

Deployment & Editions

Install in one command; run the full open-source stack; scale with Enterprise support.



cargo install

cargo install guestkit installs both the guestkit CLI and guestctl TUI binaries.

[From zero to inspecting in one line.](#)



Run from GHCR

Prebuilt public images (zyvor-ui, zyvor-api, guestkit-worker) come up via docker compose with no docker login.

[Stand up the whole console in minutes.](#)



Helm & remote deploy

A Helm chart for clusters plus scripted remote deploy for Docker hosts.

[Ship it where your fleet already lives.](#)



Full open-source stack

CLI, TUI, Python bindings, assurance APIs, web console, and KubeVirt hooks are all Apache-2.0 in the repo - Enterprise adds support, not features.

[Nothing core is withheld from the open source.](#)



Enterprise programs

SLA, air-gapped deployment packages, guided playbooks, and fleet automation for 100+ VM and regulated migrations.

[Backed help for VMware-exit programs at scale.](#)

The default eval compose stack runs without authentication - do not expose it beyond localhost. Use the production template and turn auth on before any network exposure.

Your first 30 minutes

A suggested path to value on day one.

- 1 Install**
Run `cargo install guestkit` to get the guestkit CLI and guestctl TUI, or pull the web stack from ghcr.io/hypersdk.
- 2 Score boot readiness**
`guestkit doctor vm.qcow2 --target proxmox --explain` returns a 0-100 boot assurance score with ranked blockers and root-cause chains.
- 3 Export a fix plan**
`guestkit migrate-plan vm.vmdk --target proxmox --export plan.yaml` writes an executable, reviewable migration fix plan.
- 4 Explore interactively**
`guestctl tui vm.qcow2` opens the carbon TUI with the Assurance workspace and fix-plan preview.
- 5 Gate your pipeline**
`guestkit doctor img.qcow2 --target proxmox -o json --fail-below 80` fails CI on any image that regresses below your bar.

Support & contact

GuestKit is developed by **ZyvorAI Labs**. For onboarding, a proof-of-concept, or a guided demo, contact your account team at info@zyvor.dev.

Good to know: GuestKit runs on Linux hosts and needs host tooling (`losetup`, `qemu-nbd`, and `qemu-img` for conversion); mount and in-cluster boot-inspect operations require root or a privileged pod. Offline disk analysis targets stopped VMs - running VMs return VM-spec heuristics or need the live guest agent. Deep guest intelligence is Linux-first; the native Windows agent MSI is scaffolded, and Windows relies on `virtio-win` today. LLM-assisted features require building with `--features ai`; deterministic intelligence works without it. Reporting notes: `--output` and `--export` are mutually exclusive, HTML reports truncate to 100 packages, and PDF is produced via external tools.

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